

USE CASE SPECIFICATION

1. 4 Use Case important:

- **Login:**

Field	Description
Use Case Name	Login
Actor	Admin, Lecturer, Student
Brief Description	Allows registered users to authenticate their identity and access the system based on their assigned roles.
Pre-conditions	The user must have a registered account in the system database.
Basic Flow	1. User navigates to the system's login page. 2. User enters their credentials (Username/ID and Password). 3. User clicks the "Login" button. 4. System validates the credentials against the database. 5. System redirects the User to their respective role-based dashboard.
Alternate/Exception Flow	4a. Invalid Credentials: System displays an error message ("Incorrect Username or Password") and prompts the User to retry. 4b. Account Locked: If the user fails to login multiple times, the system locks the account and displays a warning.
Post-conditions	User is successfully authenticated and logged into the system.

- **Create Attendance Session:**

Field	Description
Use Case Name	Create Attendance Session
Actor	Lecturer
Brief Description	Allows the lecturer to initialize an attendance tracking session for a specific class, generating dynamic codes for students to use.
Pre-conditions	Lecturer is logged into the system. The class schedule is assigned to the Lecturer.

Field	Description
Basic Flow	1. Lecturer selects a specific class from their dashboard. 2. Lecturer clicks "Create Attendance Session". 3. Lecturer inputs the active duration for the session (e.g., 15 minutes). 4. Lecturer confirms the creation. 5. System simultaneously generates two check-in options: a unique dynamic QR code and a 6- digit PIN 6. System displays both codes on the screen and starts the countdown timer.
Alternate/Exception Flow	3a. Missing/Invalid Duration: System prompts the Lecturer to input a valid timeframe (> 0 minutes) before proceeding.
Post-conditions	The session status is set to "Active". Students can now begin checking in.

- **Check-in:**

Field	Description
Use Case Name	Check-in
Actor	Student
Brief Description	Allows students to mark their presence in an active class session. This use case can be extended by either "Scan QR Code" or "Enter PIN" methods.
Pre-conditions	Student is logged into the system. The Lecturer's attendance session is currently "Active".
Extension Points	Input Attendance Data: The point where the student chooses the method to provide the attendance code.
Basic Flow	1. Student selects the "Check-in" function for the ongoing class. 2. System displays options to provide the attendance code. 3 [Extension Point: Input Attendance Data] The system will execute the flow of the "Scan QR Code" OR "Enter PIN" use case at this step. 4. System receives the input data (from the selected method) and verifies the validity and timestamp of the code/PIN. 5. System records the student's status as "Present". 6. System displays a success message to the student.

Field	Description
Extended Flows	3a. Extend Use Case 1 (Scan QR Code): At step 3, the student chooses "Scan QR Code". The system opens the camera, the student scans the code displayed by the Lecturer, and the decoded data is passed to step 4. 3b. Extend Use Case 2 (Enter PIN): At step 3, the student chooses "Enter PIN". The student inputs the 6-digit PIN manually and submits. The PIN data is passed to step 4. (Students choose one of these two methods to mark their attendance).
Alternate/Exception Flow	4a. Expired or Invalid Code/PIN: System displays an error ("Code expired or invalid") and denies the check-in. Status remains "Absent".
Post-conditions	The student's attendance record is successfully logged in the database.

- **Export Attendance Report:**

Field	Description
Use Case Name	Export Attendance Report
Actor	Lecturer
Brief Description	Allows the lecturer to extract attendance data of a specific class into a downloadable file format.
Pre-conditions	Lecturer is logged into the system.
Basic Flow	1. Lecturer navigates to the "Reports" module. 2. Lecturer selects a class section and specifies a date range. 3. Lecturer clicks the "Export" button. 4. System compiles the attendance data. 5. System generates a spreadsheet file (Excel/CSV format). 6. System triggers the file download to the Lecturer's device.
Alternate/Exception Flow	4a. No Data Found: If there are no attendance records for the selected filters, the system displays a "No records found" warning and stops the export process.
Post-conditions	An offline report file is successfully downloaded.

2. Remaining Use Case specifications:

- **Manage Accounts:**

Field	Description
Use Case Name	Manage accounts
Actor	Admin
Pre-conditions	Admin is logged in.
Basic Flow	Admin navigates to User Management -> Selects action (Create/Update/Delete user) -> Inputs data -> System validates and updates the database.
Alternate/Exception Flow	Admin inputs duplicated ID/Email -> System rejects and shows an error.
Post-conditions	System database reflects the updated user accounts.

- **Manage Classes:**

Field	Description
Use Case Name	Manage Classes
Actor	Admin
Pre-conditions	Admin is logged in.
Basic Flow	Admin navigates to Class Management -> Creates a new class -> Assigns Lecturer and Students -> Saves -> System links the profiles to the class.
Alternate/Exception Flow	Missing mandatory fields -> System prevents saving.
Post-conditions	Class roster is configured in the system.

- **View Real-time Attendance:**

Field	Description
Use Case Name	View Real- time Attendance
Actor	Lecturer

Field	Description
Pre-conditions	Lecturer is logged in. An attendance session is active.
Basic Flow	Lecturer opens the session -> System displays the live list of students -> System auto-updates the "Present" counter as students check in.
Alternate/Exception Flow	Network Disconnection/Sync Error: If the system loses network connection while viewing live data, it displays a warning "Connection lost, data may not be up-to-date" and automatically attempts to reconnect.
Post-conditions	Lecturer views the latest attendance metrics.

• **Update Attendance Status:**

Field	Description
Use Case Name	Update Attendance Status
Actor	Lecturer
Pre-conditions	Lecturer is logged in.
Basic Flow	Lecturer opens class list -> Searches for a student -> Manually changes status (e.g., from Absent to Present/Late) -> Saves -> System updates record.
Alternate/Exception Flow	Student Not Found: At the search step, if the Lecturer enters a Name/Student ID that does not exist in the class, the system displays the message "No matching student found". System Error on Save: If a database error occurs when the Lecturer clicks "Save", the system alerts "Update failed, please try again" and retains the previous data.
Post-conditions	Student's attendance status is manually overridden.

• **View Attendance History:**

Field	Description
Use Case Name	View Attendance History
Actor	Student

Field	Description
Pre-conditions	Student is logged in.
Basic Flow	Student accesses "My Attendance" -> Selects a course -> System retrieves and displays total absences, presences, and percentage.
Alternate/Exception Flow	No Attendance Data: If the class has not had any sessions yet or the student has no attendance records, the system displays an empty state screen with the message "No attendance data available for this course".
Post-conditions	Student views their historical data.

- **Submit Leave Request:**

Field	Description
Use Case Name	Submit Leave Request
Actor	Student
Pre-conditions	Student is logged in.
Basic Flow	Student opens Leave Request form -> Selects date and class -> Enters reason & attaches proof -> Submits -> System marks as "Pending" and notifies Lecturer.
Alternate/Exception Flow	Leaves reason blank -> System highlights field and stops submission.
Post-conditions	Leave request is logged awaiting approval.